Exercise 15.4

Solve by reducing to exact form the following equation.

1. $ydx - xdy = xydy$

Sol^n: $ydx - xdy = xydy$

$$\Rightarrow \quad \frac{dx}{x} - \frac{dy}{y} = dy$$

$$\Rightarrow \quad d\left(\log\frac{y}{x}\right) = dy$$

Integrating, we get

$$\therefore \quad \log\left(\frac{x}{y}\right) = y + c$$

2. $2xy\,dx + x^2dy = 0$

Sol^n: $2xy\,dx + x^2dy = 0 \quad \Rightarrow y \,.\, 2xdx + x^2.\, dy = 0$

$\Rightarrow \quad yd(x^2) + x^2dy = 0 \quad \Rightarrow d(x^2y) = 0$

Integrating, we get

$x^2y = c$, where c is arbitrary constant.

3. $2xydx - x^2dy = 0$

Sol^n: $y \cdot 2xdx - x^2dy = 0 \quad \Rightarrow yd(x^2) - x^2dy = 0$

Dividing by y^2, we get

$$\frac{yd(x^2) - x^2dy}{y^2} = 0 \quad \Rightarrow d\left(\frac{x^2}{y}\right) = 0$$

Integrating, we get

$$\frac{x^2}{y} = c \quad \therefore x^2 = cy$$

4. $(x + y)dy + (y - x)dx = 0$

Sol^n: $(x + y)dy + (y - x)dx = 0$

$\Rightarrow$ $xdy + ydy + ydx - xdx = 0$

$\Rightarrow$ $(xdy + ydx) + ydy - xdx = 0$

$\Rightarrow$ $d(xy) + ydy - xdx = 0$

Integrating, we get

$$xy + \frac{y^2}{2} - \frac{x^2}{2} = \frac{c}{2}$$

$\therefore$ $2xy + y^2 - x^2 = c$.

5. $xdy + (x + 1)y\,dx = 0$

Sol^n: Here,

$xdy + xydx + ydx = 0$

$\Rightarrow$ $xdy + ydx = -xydx$

$\Rightarrow$ $\frac{dy}{y} + \frac{dx}{x} = -dx$

$\Rightarrow$ $d\log(xy) = -dx$

$$\left[\because d\log(xy) = \frac{d\log(xy)}{d(xy)} \times \frac{d(xy)}{dx} \times dx = \frac{1}{xy}\left(y + x\frac{dy}{dx}\right) \times dx = \frac{dy}{y} + \frac{dx}{x}\right]$$

Integrating, we get,

$\log xy = -x + c \quad \therefore x + \log xy = c$

6. $(x^2 - ay)dx - (ax - y^2)dy = 0$

Sol^n: $x^2dx - aydx - axdy + y^2dy = 0$

$\Rightarrow$ $\frac{1}{3}dx^3 + \frac{1}{3}dy^3 - a(ydx + xdy) = 0$

$\Rightarrow$ $dx^3 + dy^3 - 3a(ydx + xdy) = 0$

$\Rightarrow$ $d(x^3 + y^3) - (3a)\,dxy = 0$

Integrating, we get

$x^3 + y^3 - 3axy = c$

7. $\sin x \cos x\,dx + \sin y \cos y\,dy = 0$

Sol^n: $\sin x \cos x\,dx + \sin y \cos y\,dy = 0$

$\Rightarrow$ $\sin x\,d\sin x + \sin y\,d\sin y = 0$

$\Rightarrow$ $\frac{1}{2}d\sin^2 x + \frac{1}{2}d\sin^2 y = 0$

Integrating, we get

$\sin^2 x + \sin^2 y = 2c$.

8. $\frac{dy}{dx} = \frac{\cos^2 y}{\sin^2 x}$

Sol^n:– $\frac{dx}{\sin^2 x} + \frac{dy}{\cos^2 y} = 0$

$\Rightarrow \quad -\operatorname{cosec}^2 x dx + \sec^2 y dy = 0$

$\Rightarrow \quad d(\cot x) - d(\tan y) = 0$

Integrating, we get

$\cot x + \tan y = c$

9. $(x + 2y - 3)dy - (2x - y + 1)dx = 0$

Sol^n: $xdy - 2ydy - 3dy - 2xdx + ydx - dx = 0$

$\Rightarrow \quad (dxy + ydx) + 2(ydy - xdx) - 3dy - dx = 0$

$\Rightarrow \quad d(xy) + d(y^2 - x^2) - 3dy - dx = 0$

Integrating, we get

$xy + y^2 - x^2 - 3y - x = c$

10. $\frac{dy}{dx} = \frac{y - x + 1}{y - x + 5}$

Sol^n: $ydy - xdy + 5dy = ydx - xdx + dx$

$\Rightarrow \quad -(xdy + ydx) + ydy + xdx + 5dy - dx = 0$

$\Rightarrow \quad -dxy + \frac{1}{2}dy^2 + \frac{1}{2}dx^2 + 5dy - dx = 0$

$\left[\because dy^2 = \frac{dy^2}{dy}dy = 2y \,.\, dy\right]$

$\Rightarrow \quad -2dxy + dy^2 + dx^2 + 10dy - 2dx = 0$

Integrating, we get

$x^2 + y^2 - 2xy + 10y - 2x = c$

11. $(x^2 + xy^2)dx + (x^2y + y^2)dy = 0$

Sol^n: $x^2dx + y^2dy + xy^2dx + x^2y\,dy = 0$

$\Rightarrow \quad d\left(\frac{x^3}{3}\right) + d\left(\frac{y^3}{3}\right) + \frac{1}{2}d(x^2 . y^2) = 0$

Integrating, we get,

$\frac{x^3}{3} + \frac{y^3}{3} + \frac{1}{2}x^2y^2 = \frac{c}{6}$ or, $2x^3 + 2y^3 + 3x^2y^2 = c.$

12. $\frac{dy}{dx} = \frac{2x + 3y + 5}{2y - 3x + 6}$

Sol^n: $(2x + 3y + 5)dx - (2y - 3x + 6)dy = 0$

$\Rightarrow \quad 2x\,dx + 3y\,dx + 5dx - 2y\,dy + 3x\,dy - 6dy = 0$

$\Rightarrow \quad 2x\,dx - 2y\,dy + 3(x\,dy + y\,dx) + 5dx - 6dy = 0$

$\Rightarrow \quad d\left(\frac{2x^2}{2}\right) - d\left(\frac{2y^2}{2}\right) + 3d(xy) + 5d(x) - 6d(y) = d(c)$

$\Rightarrow \quad d(x^2) - d(y^2) + 3d(xy) + 5d(x) - 6d(y) = d(c)$

Integrating, we get,

$x^2 - y^2 + 3xy + 5x - 6y = c$

13. $2xy^3dx - 3x^2y^2dy = 0$

Sol^n: $2xy^3dx - 3x^2y^2dy = 0$

$\Rightarrow \quad 2xy^3dx = 3x^2y^2dy$

$\Rightarrow \quad \frac{dy}{dx} = \frac{2y}{3x}$

$\Rightarrow \quad 2.\frac{1}{x}dx = 3.\frac{1}{y}dy$

$\Rightarrow \quad 2.d(\log x) = 3d(\log y) + d(c)$

Dividing on both sides by y^3, we get

Integrating, we get,

$2\log x = 3\log y + \log c$

$\Rightarrow \quad \log x^2 = \log y^3 + \log c$

$\Rightarrow \quad x^2 = cy^3.$

Hint and solution of MCQ's

1. The differential equation $Mdx + Ndy = 0$; M, N being the functions of x and y will be exact if $Mdx + Ndy = d\,f(x, y)$

2. A differential equation $Mdx + Ndy = 0$ will be exact if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

   Where, $\frac{\partial M}{\partial y}$ = partial derivative considering x constant

   $\frac{\partial N}{\partial x}$ = partial derivative considering y constant

3. All a), b) and c) are exact differential form.

4. $ydx - xdy = 0$ is not exact as it cannot directly expressible in $df(x, y) = 0$ without multiplication of I.F.

5. Since $2xydx - x^2dy = 0$

   $\Rightarrow \quad \frac{2xydx - x^2dy}{y^2} = 0 \left(\text{multiplying } \frac{1}{y^2}\right)$

   $\Rightarrow \quad d\left(\frac{x^2}{y}\right) = 0$

   $\therefore$ The integrating factor is $\frac{1}{y^2}$

6. Since first order differential equation of the form $Mdx + Ndy = 0$ is exact if $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$.

   We have $x^2\,dx + y^2\,dy = 0$, where $M = x^2$ and $N = y^2$such that $\frac{\partial M}{\partial y} = 0$ and $\frac{\partial N}{\partial x} = 0$

   $\therefore \quad x^2\,dx + y^2\,dy = 0$ is exact.

7. $\sec^2x\,dx + \text{cosec}^2y\,dy = 0$

   Integrating,

   $\Rightarrow \quad \tan x - \cot x = C$

8. $(x + y)dy - (x - y)dx = 0$

$\Rightarrow \quad xdx + ydy - xdx + ydx = 0$

$\Rightarrow \quad (xdy + ydx) + \frac{1}{2}d(y^2) - \frac{1}{2}d(x^2) = 0$

$\Rightarrow \quad d(xy) + \frac{1}{2}d(y^2) - \frac{1}{2}d(x^2) = 0$

On integration,

$$xy + \frac{y^2}{2} - \frac{x^2}{2} = 0$$

$\Rightarrow \quad 2xy - x^2 + y^2 = 0$